

Roll No. ....

**DEEM-01**

**B. TECH. (ME)**  
**(FIFTH SEMESTER)**  
**MID SEMESTER EXAMINATION, Oct., 2022**  
**ENGINEERING ECONOMICS AND PLANNING (Elective)**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you mean by Elasticity ? Explain Demand Elasticity and Supply Elasticity with an example and a neat diagram. (CO1)

OR

- (b) Explain Demand and Supply in Economics. Draw a typical demand-supply diagram and explain market equilibrium. (CO1)

2. (a) What do you mean by Engineering Economics ? Explain Microeconomics and Macroeconomics. (CO1)

OR

- (b) Explain flow in economy. What do you mean by circular economy ? (CO1)

**P. T. O.**

3. (a) Suppose that demand and supply of apples are described by the following equations : (CO1, CO2)

$$P = 100 - 3Q \text{ (demand)}$$

$$P = 20 + Q \text{ (supply)}$$

- (i) Calculate the equilibrium price and quantity.  
(ii) Suppose a \$4 tax is placed on apples. What is the new equilibrium quantity ?

How much do consumers pay to get this quantity ? How much do suppliers receive for selling the quantity ?

OR

- (b) You are the manager of a store that carries generic soft drinks. Due to a local economic boom, your customer's incomes are forecasted to rise by 5% during the next month. The income elasticity of demand for these products is estimated to be - 2.0. Estimate the change in the quantity of your soft drink orders required to accommodate the new demand without a surplus or shortage inventory. (CO1, CO2)

4. (a) Explain the terms : Direct material cost, direct labour cost, overheads total cost. (CO2)

OR

- (b) Explain the law of returns, giving reasons for increasing returns, constant returns and decreasing returns. (CO3)

(3)

5. (a) The various cost components for production of 30000 units per annum are given : Direct materials ₹ 6 per unit, direct labour ₹ 5 per unit, fixed overheads ₹ 60,000, variable overheads ₹ 2.50 per unit. Find the total cost and total cost/unit. (CO2)

OR

- (b) Find the future sum of ₹ 10,000 after one year at an interest rate of 10% if the interest is compounded : annually, half yearly, quarterly and monthly. (CO3)